

Rebuttal for CoRL Submission #2468

We thank the Area Chair and reviewers. We address the requested progress baselines, loss optimization, mask scope, component ablations, and evaluation protocol below.

[AC, R1, R3] Is VPCE only task progress, and how does it compare with phase/hierarchical conditioning? We thank the reviewers and AC for highlighting this valuable comparison. We added phase-4 and normalized progress-20 predictors on the same held-out trajectories. Table I shows that VPCE consistently improves contact-prediction AUC/F1 over both alternatives. Thus trajectory time explains part of contact timing, but current wrist RGB and robot motion contain additional contact-specific information.

TABLE I
CONTACT-PREDICTION AUC/F1.

Task	Phase	Prog.	VPCE
SWIPE	.670/.000	.853/.597	.963/.759
WIPE	.666/.000	.743/.406	.885/.705
INSERT	.684/.000	.808/.539	.946/.735

line with a state-dependent 15-step horizon.

[R1, R3] Does the teacher transfer, and how much target tactile data is required? The source VPCE teacher is trained on a source dataset containing 639 tactile trajectories (545 train / 94 held-out). For downstream target-domain training, we use 60 tactile trajectories each for WIPE/SWIPE/INSERT, respectively. On trajectory-disjoint held-out data, the target-domain teachers obtain AUC/F1 .912/.731, .960/.774, and .940/.725. Thus the source teacher covers diverse contact regimes, while 60 target trajectories suffice to generalize to unseen trajectories in each downstream domain.

[R2] Are the learned loss weights well posed? The weighting was designed to emphasize sparse, control-critical contact transitions. However, optimizing γ and δ through the weighted denoising loss lets the model reduce the objective by lowering the weights. At $\bar{c}=1$, checkpoint weights are 1.005/2.1/4.4 on SWIPE/WIPE/INSERT, confirming task-dependent behavior. We replace them with $w(\bar{c}) = \text{clip}(1 + \alpha\bar{c}, 1, 1 + \alpha)$, where fixed α receives no gradients; retrained-policy experiments validate this schedule.

[R2] Does Eq. (9) mask proprioception, and how is free-space motion smooth? The reviewer is correct that the paper’s notation and original post-encoder routing conflated proprioception, gripper width, and tactile features. Although input-level dropout targeted tactile keys, the post-encoder mask could zero the entire low-dimensional tail. We corrected the routing to always retain robot state and gripper width and mask only tactile/contact features. This controlled variant obtains 13/20 on SWIPE and produces more coherent free-space approach motion. Action-chunk prediction and receding-horizon execution provide additional temporal continuity. We will revise Eq. (9) to distinguish f_{prop} from f_{tac} .

[AC, R3] Which components matter, and why retain gating? We will move the component ablations from the

appendix to the main text. They show that horizon conditioning and contact-adaptive tactile masking are critical to success. Although gating does not improve binary success, the supplementary video shows that it enables more direct and stable contact alignment, whereas the no-gating policy requires more corrective inference and actions. Thus, gating improves trajectory quality by adaptively balancing visual and low-dimensional feedback.

[R2] Could operator-held props assist the policy? After rollout start, the holder did not track the end effector, reposition the object, or assist alignment; all methods used the same placement, success criterion, and 60-s timeout. This shared protocol did not rescue non-VPCE baselines on SWIPE: DP-V and DP-VT achieved 1/20 and 2/20, whereas VPCE-DP-V and VPCE-DP achieved 17/20 and 18/20. Thus, the held setup alone does not explain the gap; the predicted contact horizon helps the policy commit through final approach and contact. To reduce sampling uncertainty, we expanded VPCE-DP to 30 trials/task, obtaining 28/30 on SWIPE (93.3%), 22/30 on WIPE (73.3%), and 18/30 on INSERT (60.0%).

[R2, R3] Why exclude tactile input from VPCE, and how are labels and contact directions handled? VPCE intentionally excludes tactile input because its role is to anticipate contact before touch is informative and to permit tactile-free deployment; tactile is used as privileged training supervision. In the full policy, tactile feedback remains available after contact. For the current planar tasks, contact labels use the temporal change of a base-frame z -force proxy. A threshold of 0.095 is used for the source, SWIPE, and INSERT data; WIPE uses 0.0507 due to its different sensor scale. Each threshold is selected once from the training force-change distribution and then fixed. A $0.6\times-1.4\times$ threshold sweep changes the 15-step positive-label rate smoothly (SWIPE .296 $\rightarrow$.217, WIPE .582 $\rightarrow$.379, INSERT .390 $\rightarrow$.290), rather than creating labels only at one tuned value. Contacts with changing normals can instead use $\|\Delta\mathbf{F}\|_2$, local normal/shear changes, hysteresis, or learned change-point labels without changing the VPCE-policy interface.

[R1, R3] What is the main contribution relative to predictive force guidance and tactile representation learning? We will sharpen the claim: the contribution is the identification of pre-contact stall, tactile-supervised distillation of a compact future-contact horizon into visual-proprioceptive inputs, and evidence that this signal can initiate contact even without tactile sensing at deployment. The individual gates and loss weights are not claimed as the primary novelty. Unstructured tactile encoders such as Sparsh are complementary for representing post-contact feedback; comparing them does not replace the question addressed here—how to act before tactile observations become informative. We will expand related work and state the missing Sparsh/full hierarchical policy comparisons as evaluation limitations.